4.1. Notes

Site: Eduvos LMS
Course: ITNMA4 (First Block)
Book: 4.1. Notes
Printed by: Sibabalwe Mfisa
Date: Wednesday, 15 July 2026, 6:14 PM

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1. Learning outcomes

By the end of this lesson, you should be able to:

- Examine 5G concepts and architecture.

Prescribed Reading

Shyam Varan Nath, Ananya Simlai& Oğuzhan Kara. Mastering 5G Network Design, Implementation, and Operations: A comprehensive guide to understanding, designing, deploying, and managing 5G networks. 1st Edition. 2023. Packet Publishing.

Chapter 2

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2. 5G Network Concepts

A 5G network system is made up of different Network Functions (NF) which perform different activities. The NFs are connected together to achieve their intended function.

The NFs have been designed to operate over Commercial Off-The-Shelf (COTS) hardware and are no longer complicated or expensive because the hardware is virtualized.

2.1. Access and Mobility Management Function (AMF)

Access and Mobility Management Function (AMF) in 5G is a critical component of the core network responsible for managing user access, mobility, and connectivity. It ensures a seamless and secure experience for users as they connect to and move within the 5G network.

Session Management Function

Session Management Function (SMF) -The SMF is a key network function within the 5G core network (5GC) that plays a crucial role in managing and establishing sessions for user equipment (UE)

User Plane Function (UPF)

User Plane Function (UPF)- The UPF is a crucial component within the 5G core network (5GC) responsible for handling the user data plane. It manages the efficient and secure transport of user data between the user equipment (UE) and external networks.

NOTE : Refer to the prescribed resources for all the other NFs

2.2. PDU Sessions

PDU (Packet Data Unit) session refers to the establishment of a communication session for packet-switched data between a user equipment (UE) and the 5G core network (5GC).

- PDU sessions play a key role in facilitating the transfer of user data, supporting various services and applications.
- They are a key component in supporting diverse applications and services with varying quality of service requirements.

2.3. Quality of Service (QoS) in 5G (fifth-generation)

- QoS refers to the set of parameters and mechanisms that ensure the delivery of reliable, efficient, and predictable communication services.
- QoS is crucial for supporting a diverse range of applications and services with varying requirements in terms of latency, throughput, reliability, and other performance metrics.
- 5G QoS is essential for delivering a reliable and efficient wireless communication experience.
- Key aspects include End-to-End QoS, Differentiated Services, and Latency Management.

2.4. Session and Service Continuity (SSC)

- (SSC) in 5G is a set of capabilities that enable uninterrupted connectivity and service delivery for users as they move within the network or experience handovers between different technologies.
- It plays a crucial role in ensuring a seamless and reliable communication experience in dynamic and evolving wireless environments.

2.5. Activity

Activity

Discuss the network functions that are connected to the AMF as shown below;

